Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/20/2026
PI Name:	Gerecht, Sharon Ph.D.		
Project Title:	Creating Non-Human Tended Microphysiological Systems (PI: Gerecht)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21218	**Congressional District:**	4
Comments:	Moved to Duke in 2022 (previously at Johns Hopkins University)		
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH Directed Studies and Initiatives
Start Date:	02/01/2024	**End Date:**	03/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2027 per TRISH (Ed., 12/10/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-SNT0101		
Performance Goal No.:			
Performance Goal Text:			
	Microphysiological systems (MPS), also known as organs-on-chips, are engineered to recapitulate organ-level and systemic functions in the context of human patho/physiology. Recent studies are increasingly demonstrating the potential of MPS to model the impact of space stressors (i.e., radiation, microgravity, oxidative stress) on astronauts' health, towards identifying the health risks and developing effective countermeasures. The current state of the art in the design and utilization of MPS for space research involves two major unmet needs: (i) rigorous validation of their physiological relevance, and (ii) compatibility with manned and unmanned flight missions. We propose to address these two needs by engineering the next-generation human MPS platform that will combine high biological fidelity of experimental conditions (emulating human physiology on Earth and in space) with configurable design and automated operation (for broad and high impact on addressing the effects of space stressors). To this end, we will focus on		

Task Description:	recapitulating human physiology in studies of tissue systems that are of highest interest for spaceflight: in situ sensing/monitoring, and automated operation. We will focus on four organ systems that are of the highest interest for space research because of their documented adverse effects on astronauts: bone marrow (BM); cardiac muscle (CM); small artery; and blood-retinal barrier (BRB); individually, and in combination. To enable individualized ("astronaut on chip") physiological studies, all tissues will be derived from the same human induced pluripotent stem cells (iPSCs) and matured to recapitulate organ-level functions. To enable space studies, with and without crew support, we will design the platform for compatibility with automated operation, with in situ sensing, data acquisition, and analysis. To validate the platform, we will demonstrate its capability to recapitulate human physiology on Earth. To study the physiological responses to spaceflight related stressors, we will engineer all tissues from iPSCs gene-edited to systemically express oxidative stress, a known key stressor in space, and compare to unedited control tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our previous work on single- and multi-organ MPS platforms has a track record of innovative designs and the organ-level and systemic studies of human physiology. Uniquely, we engineered patient-specific tissues that are matured to adult-like phenotypes and maintained for weeks and months while allowing their communication via vascular circulation. This capability was established by maintaining the individual tissue/organ compartments while linking them by vascular flow to allow cross-talk, as in our body, and will now be advanced by the following contributions: (1) Gene-edited iPSCs expressing high oxidative stress, a widely recognized space stressor; (2) Incorporation of vascular flow, endothelial barriers, and circulating immune and hematopoietic cells generated by iPSC-derived bone marrow; (3) Implementation of automated techniques for tissue culture, in situ sensing and analysis; (4) Systematic validation of engineered tissues, individually and in the integrated platform, against their native counterparts; (5) Platform design enabling the transition to non-human-tended flight hardware. The studies will specifically address the key risks identified by the NASA Human Research Program (HRP) (CV-202, CV-101, CV-102, Degen-1, 2, 3, IM-101, Pharm-601) in an "astronaut on chip" setting. We will genetically induce oxidative stress into iPSCs, interrogate the resulting changes in multiple tissues, identify targeted pathways, and test protective countermeasures (CV-202). Furthermore, we will design the platform to enable an automated operation and transition to the flight hardware to enable the study of tissue responses to space stressors during crewed and uncrewed missions (CV-101; CV-102; Degen-1, 2, 3; IM-101). Finally, non-human tended MPS represent an open, standardized, and configurable platform for developing countermeasures to the effects of space environment (e.g., pharmacokinetics, pharmacodynamics, drug testing) (Pharm-601). As we are recapitulating the four organ systems of the highest interest to space research (bone marrow, cardiac muscle, small artery, and inner blood-retinal barrier), due to their adverse effects, our novel MPS platform will allow us to assess health risks during space travel and elucidate physiological and pathological mechanisms underlying spaceflight. This will ultimately advance precision medicine and accelerate the development of protective countermeasures.
Task Progress:	In the first year of this project, we have made progress toward each of these aims and have accomplished the planned milestones and deliverables: Aim 1: 1.1 Engineering of adult-like human tissues (heart, bone marrow, vasculature, blood-retinal barrier) using the same iPSC line 1.2 Cultivation of individual tissues with vascular flow 1.3 Introducing circulating immune cells or bone marrow (in progress) Aim 2: 2.1 Knockdown Nrf2 (and/or downstream HMOX-1) to regulate oxidative stress response to mimic space flight responses Aim 3: 3.1 Early inputs from flight hardware contractors 3.2 Design specs: modular and configurable platform in a 96-well plate footprint SUMMARY OF ACCOMPLISHMENTS In Aim 1, we developed functional 3-dimensional models of the blood-retina barrier, small artery, cardiac muscle and bone marrow, starting from human iPSCs. For all four tissues, we confirmed their genotypic and phenotypic integrity by the expression of canonical cell-specific protein markers, physiological morphological and functional properties. Comprehensive characterization confimed that the engineered tissues robustly recapitulate the organ level readouts of their native counterparts. In Aim 2, our team at Duke University has obtained from another lab the matching wild type (WT) iPSCs and iPSCs with genetically induced oxidative stress via a knockout (KO) of antioxidant gene Nrf2, and differentiated both lines into arterial iECs to compare their differentiation efficacy and phenotype. Our team at Columbia University has gene-edited our standard, most frequently used line of iPSCs to induce oxidative stress via a knockout (KO) of antioxidant gene Nrf2, and tested the ability of the wild-tye and edited lines to differentiate into cardiac and vascular cells. In Aim 3, we obtained early inputs from Dr Lucy Low, Chief Scientist at Axiom Space and Dr Stefanie Countryman, Director of BioServe Space for how to design of our final MPS platform for eventual transition into the flight hardware. To establish the design specs for the planned modular and configurable platform in a 96-well plate footprint, started the design and testing of simplified yet functional tissue models suitable for uncrewed space studies. PLANS FOR YEAR 2 In year 2, the work on Aim 1 will focus on the completion of the following three milestones:

	1.3 Integration of multi-tissue platforms; 1.4 Introducing circulating immune cells or bone marrow; and 1.5 Cross-validation of experimental results between the two labs. In parallel, we plan to make major progress with the milestone 1.6: Developing in situ sensing modalities (cell reporters, cell tracking, barrier permeability, contractility, signal propagation). The work on Aim 2 will focus on making major progress on milestones: 2.2 Study tissues from gene-KO iPSCs vs controls; and 2.3 Testing of countermeasures. We will incorporate the tissues derived from the Nrf2 knockdown cells into the MPS to evaluate the effects of oxidative stress, towards establishing a multi-tissue platform that can recapitulate the cumulative effects of spaceflight on human organs and is designed to facilitate translation into spaceflight hardware. The work on Aim 3 will focus on milestones: 3.2 Design specs: modular and configurable platform in a 96-well plate footprint; and 3.3 Defining capabilities for automated operation and data analysis. This approach will enable new insights into the pathophysiological mechanisms and therapeutic targets associated with spaceflight, to form basis for studies in year 3 and beyond on discovering and testing protective countermeasures for long-term missions.
Bibliography Type:	Description: (Last Updated: 05/29/2026)
Articles in Peer-reviewed Journals	Lin Y-Y, Warren E, Macklin BL, Ramirez L, Gerecht S. "Endothelial–pericyte interactions regulate angiogenesis via VEGFR2 signaling during retinal development and disease." Invest Ophthalmol Vis Sci. 2025 Sep;66(12):45. https://doi.org/10.1167/iovs.66.12.45 ; PMID: 40970668; PMCID: PMC12453065 , Sep-2025